Lessons

Graphing the Solution to an Inequality

You have already used four symbols to compare numbers.

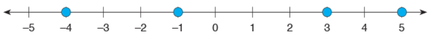
$5 > 3$ means 5 is greater than 3.

$-4 < -1$ means -4 is less than -1.

$5 \geq -1$ means 5 is greater than or equal to -1.

$-1 \leq 3$ means -1 is less than or equal to 3.

You can easily determine that these statements are true. Compare their locations on the number line. Values to the right are greater than values to the left.



When an inequality contains a variable, the variable represents a range of values.

Think about the statement $x < 3$. The variable x represents a value that is less than 3. There are many values that will make the statement true. The variable could represent 2, 0, −50, or even a fractional quantity such as $\frac{1}{2}$ or 1.75.

The **solution set** of an inequality is the set of all the values that can make the inequality true. One way to show a solution set is to graph it on a number line.

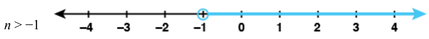
Example 1

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1. Draw a circle on the number line to represent the number in the inequality. Draw an *open circle* if the number is not a solution. Draw a *closed circle* if the number is a solution.
2. Darken the line and the arrow to show that all the values in that direction are solutions.

Example	Graph	Explanation
$x < 4$		4 is not a solution.
$x > -1$		−1 is not a solution.
$x \geq 0$		0 is a solution.
$x \leq 3$		3 is a solution.

An inequality has a range of solutions. When an inequality has only one variable, its solution set can be graphed on a line. In the example below, an open circle shows that the graph starts at -1 but does not include -1. The shaded portion indicates that all the numbers greater than -1 are solutions to the inequality.



A **linear inequality** has two variables. Study the example below to learn how to graph a linear inequality.

Example 2

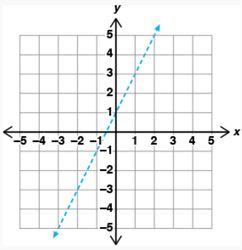
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Graph the solution set to the inequality $y > 2x + 1$.

Step 1

Graph the inequality as though it were a linear equation. Draw the line on the coordinate plane.

- Use a solid line for the symbols ≤ and ≥.
- Use a dashed line for the symbols < and >.



Step 2

The dotted line divides the graph into two regions. One of the regions contains the solutions to the inequality. Pick a point on one side of the line and test it in the inequality. The origin (0, 0) is often a good choice.

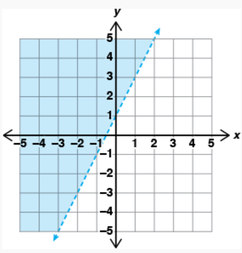
$y = 2x + 1$

$0 = 2(0) + 1$

$0 = 1$ False

Point (0, 0) is not a solution.

Shade the region on the opposite side of the line.



Any point in the shaded region is a solution to the inequality. Points that lie on the line are not a solution because the line is drawn with dashes.

If you are given a graph of an inequality, you can tell whether a point is a solution to that inequality. Remember, if the line is solid, the points on the line are also solutions.

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Lesson 5: Inequalities

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